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Design and implementation of a novel interior permanent magnet bearingless slice motor

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Abstract

In this paper, we present a bearingless motor with a novel segmented dipole interior permanent magnet (IPM) slice rotor. The segmented dipole IPM rotor contains a unique pattern of interior permanent magnets arranged to generate a dipole air gap flux pattern. The magnets are encapsulated within an electrical steel rotor structure. The stator contains a three-phase, four-pole winding for suspension and a three-phase, two-pole winding for rotation. We present analyses of several candidate rotor designs. The analyses indicate that the segmented dipole IPM rotor achieves a reduced trade-off between force and torque capacity and relatively symmetric force dynamics as compared to prior art designs and alternate topologies. Symmetric and decoupled force dynamics allow a simple force decoupling algorithm to be used. We designed, constructed, and tested a prototype system. We experimentally demonstrate that the prototype system can achieve stable levitation and open-loop rotation.

Index Terms—

bearingless motor; interior permanent magnet; magnetic levitation

I. Introduction

A bearingless motor combines the function of a magnetic bearing and electric motor to both levitate and rotate a rotor. This enables contact-free operation, which is advantageous in applications requiring low friction, long operational lifetime, and high purity or cleanliness

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[1]. When the rotor's length-to-diameter ratio is relatively small, a bearingless motor can passively stabilize the rotor's axial translation and out-of-plane tilts. Therefore, active stabilization via feedback control is only needed for two radial translations. Such a construction, i.e., slice motor, effectively reduces the number of required sensors and power electronics, simplifying the overall control system [2], [3]. This makes bearingless slice motors attractive for some special cases, e.g., blood pumps and semiconductor industry applications [4], [5]. Various types of bearingless slice motors have been developed over the last 20 years [6]. These includes surface permanent magnet (SPM) [7], consequent pole [8], homopolar [9]–[11], hysteresis [11], [12], synchronous reluctance [13], and flux-switching bearingless slice motors [14].

In this paper, we present the design, construction, and testing of a novel dipole interior permanent magnet (IPM) bearingless slice motor. This is targeted for use as a blood pump in extracorporeal life support (ECLS) applications, which is the subject of the first author's Master's thesis [15]. The design of SPM and IPM bearingless motors with p pole rotors and p–2 pole suspension windings generally presents a trade-off between force and torque capacity [1, pp. 210]. The trade-off arises because the magnet thickness affects the force and torque capacities differently. That is, thick magnets are desirable to generate a large air gap flux and improve the motor's torque capacity, i.e., the torque generated with the maximum allowable winding current. However, thicker magnets result in a higher reluctance path for the suspension flux, reducing the force capacity. Prior art bearingless IPM designs have used thin, shallowly buried permanent magnets to achieve high suspension force generation [16]. However, the improved force capacity comes at the expense of torque performance. Other work has addressed this trade-off by burying thick permanent magnets deeply within the rotor iron [17]. However, this solution results in asymmetric suspension force production that must be accommodated with asymmetric suspension current commands.

To overcome this problematic trade-off, we have designed a dipole IPM rotor as shown in Fig. 1a. This design, which we refer to as a segmented dipole IPM rotor, contains a simple dipole magnet arrangement yet exhibits a reduced trade-off between force and torque capacity as compared to alternate topologies. In addition, our rotor design exhibits relatively symmetric radial suspension dynamics, which allows us to use a simple decoupling algorithm and single-input single-output (SISO) controllers for the levitation control. The design is also simple to manufacture. We have constructed and experimentally characterized our dipole IPM rotor.

The rest of the article is organized as follows: Section II explains the operating principle. Section III presents the design and construction of a prototype system. Section IV explains the control and instrumentation. Section V presents the test results. Section VI concludes the article and suggests further work.

II. Operating Principle

A rendering of the segmented dipole IPM bearingless motor magnetic design is shown in Fig. 2. The design consists of a segmented dipole IPM slice rotor and a twelve tooth stator. The rotor contains a pattern of permanent magnets embedded within a stack of electrical steel laminations, where the magnets are arranged to generate a dipole (two-pole) flux pattern in the motor air gap. The stator is a temple configuration [2] with a separated winding scheme that consists of two three-phase windings sets: a two-pole rotation winding, and a four-pole suspension winding. In the remainder of this section, we describe the suspension and rotation operating principles of the prototype motor.

A. Suspension

The suspension force operating principle for our motor is typical of bearingless slice motors as described in [2]. The dipole rotor generates a two-pole ($n_{p_r} = 2$) bias flux in the air gap. The bias flux provides passive stability in the axial (z) and tilt (θ_x, θ_y) directions, shown in Fig. 2, due to restoring reluctance forces that are generated when the rotor is perturbed in these degrees of freedom. The radial degrees of freedom (x, y) are open-loop unstable due to the negative stiffness associated with the bias flux and must be actively controlled to achieve stable levitation. Radial forces on the rotor are produced by generating a four-pole ($n_{p_s} = 4$) suspension flux in the air gap. This follows the $n_{p_s} = n_{p_r} + 2$ suspension force principle discussed in [1], where the rotor pole number is $n_{p_r} = 2$ and the suspension flux pole number is $n_{p_s} = 4$.

The radial suspension force principle for the segmented dipole IPM rotor is illustrated in Fig. 3. For simplicity in illustrating the principle, the stator is shown with a two-phase, four-pole suspension winding, rather than the three-phase winding in the actual construction. The interaction of the four-pole suspension flux with the two-pole rotor bias flux produces radial forces on the rotor by decreasing the net flux density on one side of the rotor and increasing it on the other. The orientation of the rotor bias flux pattern with respect to the stator fixed suspension windings will change as a function of the rotor angle ϕ. To accommodate this, a force commutation algorithm is generally required to decouple the force generation from the rotor angle. The details of the commutation and control algorithms for our prototype system will be discussed in Section IV-C.

It can be seen from Fig. 3 that the reluctance paths for the four-pole suspension flux are not equivalent for the x and y force generation. This feature, which results in asymmetric radial force production, is common to other salient-pole dipole rotors as described in [1, pp. 131–132]. The asymmetric force production can be accommodated in the force commutation algorithms but at the expense of control complexity. In the case of our segmented dipole IPM rotor, the asymmetry is sufficiently small such that stable operation was demonstrated with no added complexity to the commutation algorithms.

B. Rotation

The rotation principle for the IPM bearingless motor is the same as for a conventional non-bearingless IPM motor. Torque on the rotor is produced by generating a two-pole rotation magnetomotive force (MMF) in the air gap to interact with the two-pole bias flux. A field oriented control (FOC) commutation scheme is used where the rotation winding current is commanded in a rotor-fixed frame. Two orthogonal axes are defined in the rotor-fixed dq-frame: a direct-axis (d-axis) along the rotor dipole axis of magnetization, and quadrature axis (q-axis) advanced 90 electrical degrees from the direct axis. The torque production equation for an IPM is given in [18] as

$$\tau = qp\left(\lambda_{PM} - \left(L_q - L_d\right)i_d\right)i_q\,. \tag{1}$$

Here, q is the number of rotation winding phases, $p = n_{p_r}/2$ is the number of rotor pole pairs, λ_{PM} is the permanent magnet flux linkage along the d-axis, L_d is the d-axis inductance, L_q is the q-axis inductance, i_d is the commanded d-axis current, and i_q is the commanded q-axis current. The segmented dipole IPM rotor is designed with $L_q > L_d$ such that permanent magnet and reluctance torque can simultaneously be generated with positive q-axis current and negative d-axis current. To simplify the torque commutation in our intitial tests, the d-axis current is set to zero and only q-axis current is commanded. With $i_d = 0$, the motor behaves like a synchronous permanent magnet (PM) machine.

III. Design and Construction

In this section we describe the design and construction of the prototype segmented dipole IPM slice rotor and bearingless motor system. A photograph of the prototype system is shown in Fig. 1b. We begin with an overview of the approach taken during the design phase, and then discuss the details of the rotor, stator, and windings.

A. Design Approach

A heuristic, iterative design process was used to develop the segmented dipole IPM rotor. We developed preliminary concepts, analyzed them with FEA, and formulated new concepts and design iterations based on the analysis results. The rotor design footprint was constrained to 50 mm diameter by 10 mm height for compatibility with an existing prototype stator [10]. We focus on two-pole IPM rotor designs with four-pole suspension ($n_{p_r} = 2$, $n_{p_s} = 4$) due to their less severe force vs. torque capacity trade-off as compared to surface permanent magnet bearingless motors and motors with the $n_{p_s} = n_{p_r} - 2$ configuration [1, pp. 210].

A total of 13 candidate designs, 2B through 2N in Fig. 4, were evaluated using 2D finite element analysis (FEA) performed in the open source software FEMM [19]. In addition to these two-pole IPM designs, a simple disc dipole PM design (2A) and two four-pole IPM designs (4A and 4B) were evaluated for comparison. For each design, simulations were performed to calculate the x and y force constants, x and y negative stiffnesses, and torque constant. The results of these simulations are shown graphically in Fig. 4. Here, Fig. 4a shows the torque constant vs. force constant, Fig. 4b shows the x force constant vs. y force constant, and Fig. 4c shows the x negative stiffness vs. y negative stiffness. In these plots, the force and torque constants are normalized by the number of turns in the prototype stator [10] suspension and rotation winding sets. The stator design is further discussed in Section III-D. In general, we see that the two-pole IPM designs (2B through 2N) outperform the others in terms of simultaneously achieving large force and torque capacities. This is because in the two-pole IPM rotor, the entirety of the stator-generated suspension flux does not pass through the magnets as it invariably does in four-pole IPM and two-pole PM rotors.

Thus, thicker magnets can be employed to increase torque capacity with less effect on force capacity. Ultimately, design 2N, which we refer to as the segmented dipole IPM rotor, was chosen to manufacture and test. Of the candidate designs, this design exhibits the best balance of symmetric force capacities, symmetric negative stiffnesses, balanced force and torque capacity, and manufacturability.

Additional simulations of design 2N were performed to characterize the predicted performance. Fig. 5 shows three air-gap flux density distributions that contribute to the symmetric radial suspension force and negative stiffness characteristics of design 2N. In Fig. 5a, the air-gap flux produced by the permanent magnets is shown to approximate a sinusoidal distribution with a dominant 1x spatial harmonic component. In Fig. 5b and Fig. 5c, the air-gap flux produced by the suspension windings approximate sinusoidal distributions with dominant 2x spatial harmonics. We attribute the symmetry in this design in part to the limited higher spatial harmonics in the permanent magnet and suspension flux distributions. We also attribute the symmetry in the suspension force dynamics to the similar magnitude 2x harmonic components for the x-axis and y-axis suspension flux distributions.

In Fig. 6, the rotor mechanical angle ϕ is swept from 0 to π and several bearingless motor performance parameters are calculated at each angle using the FEMM simulation. Fig. 6a and Fig. 6b show the radial forces with 1A x-axis and y-axis current commands as a function of rotor angle. These results indicate there is some level of force cross-coupling, as a y-axis force command generates some level of x-axis force and vice-versa. The commanded forces also vary with rotor angle. This can be attributed to the interaction of the rotor saliency, stator saliency, and higher harmonics in the stator and rotor generated air-gap flux. Fig. 6c shows the simulated x-axis and y-axis negative stiffnesses as a function angle. These parameters also show a rotor angle dependence, which can be attributed to the rotor saliency, stator saliency, and rotor air-gap bias flux pattern. Finally, Fig. 6d shows the rotor torque as a function of both mechanical angle and commutation angle, i.e., the relative angle of the rotation MMF with respect to the rotor mechanical angle. The total rotation current ($i_r = \sqrt{i_d^2 + i_q^2}$) is held constant at 1 A. The predicted ripple torque is fairly large, with the ripple ratio at a 90° commutation angle (pure q-axis current) equal to 96.1%. At this current level, the torque is dominated by cogging torque. The large cogging torque in this design is attributed to the stator saliency and strong permanent magnets.

B. Rotor

An annotated diagram of the segmented dipole IPM slice rotor prototype is shown in Fig. 2b. The rotor is constructed from a stack of electrical steel laminations which were laser cut from 24 Gauge (0.635 mm) M-19 electrical steel sheets coated in C5 insulating varnish. The laser cut laminations are stacked and bonded using Remisol EB 548 bonding varnish. The rotor stack diameter is 50 mm and the height is 10 mm. The rotor stack contains slot features to accept the permanent magnets. Relief cuts are added at the corners of each slot to prevent interference between the magnet and slot corners. The minimum material width was kept to 0.75 mm at the saturating bridge locations for structural and manufacturing reasons. Three magnets are inserted into each of the four slots on the rotor and epoxied in place using E-30CL 2-part epoxy from Loctite. Using three magnets per slot allows for off-the-shelf permanent magnets rather than more expensive custom size magnets. The magnets are nickel-coated N48 grade NdFeB magnets from Apex Magnets (quantity six M20X10X2MMBL and quantity six M10X5X2MMBL). A central bore is also included in the lamination stack for fixturing during calibration and testing of the rotor.

C. Stator

The stator structure used for our motor is the same as that developed in [10] and shown in Fig. 2. The stator is made up of twelve L-shaped teeth, a bottom plate, 36 coils and a housing structure. The teeth and bottom plate are wire-EDM machined from stacks of 0.35 mm electrical steel (M330-35A). Three concentrated coils of 21 AWG magnet wire are installed on each stator tooth, and the teeth are installed on the base in a temple configuration.

D. Windings

The three-phase, four-pole suspension winding and three-phase, two-pole rotation winding are realized by connecting the stator coils external to the stator via terminal blocks. A diagram of the coils used in each winding set is shown in Fig. 7. The two-pole rotation winding set consists of two layers of coils. Each coil in the bottom layer contains 140 turns and each coil in the top layer contains 52 turns. The eight coils in each phase are connected in series, and the three phases (U, V, and W) are connected in a wye-configuration. The phases are arranged spatially such that the U-phase magnetic axis is aligned with the x-axis and the V-phase and W-phase magnetic axes are ±120 mechanical degrees from the x-axis. The number of turns and distribution between the layers was determined to minimize the total harmonic distortion of the two-pole MMF [10].

The four-pole suspension winding set consists of one layer of coils each with 108 turns. The four coils in each phase are connected in series such that the direction of current flow alternates in each adjacent coil. This creates a four-pole MMF when excited. The three phases (A, B, and C) are connected in a wye-configuration. The phases are arranged such that the A-phase magnetic axis is aligned with stator tooth 1, 15 mechanical degrees advanced from the x-axis. The B-phase and C-phase magnetic axes are oriented ±60 mechanical degrees (±120 electrical degrees) from the A-axis. This winding set was constructed by re-terminating the eight-pole, 108 turn-per-coil winding described in [10], [11].

IV. Control System

In this section, we describe the design and implementation of the control system for our bearingless motor prototype. Closed-loop feedback control of the rotor's radial degrees of freedom (x and y) is required due to the negative stiffness that arises from the bias flux as discussed in Section II-A. As for the rotation, we present an open-loop torque control scheme using field-oriented control for commutation. In designing the suspension and rotation control systems, we assume the force and torque generation in the motor are decoupled. Block diagrams for the rotation and suspension systems are shown in Fig. 8. The control system includes angle and position sensors, power amplifiers, control and commutation algorithms, and controller hardware. In the remainder of this section we describe the details of each of the control system subcomponents.

A. Sensors

The sensing system consists of four optical sensors, 12 Hall elements, and associated conditioning electronics. A diagram of the sensor configuration is shown in Fig. 9. The four optical sensors are used to measure the rotor radial position for suspension feedback control. A piece of white tape is placed on the rotor outer diameter to increase the reflectivity. The 12 Hall elements are used to measure the the rotor mechanical angle for force and torque commutation. The Hall elements are also used to measure the rotor axial and tilt motions. These additional measurements, while not required for control as the tilt and axial degrees of freedom are passively stable, are useful for characterization and diagnostic measurements in a prototype system. The detailed design and evaluation of the sensing system is presented in our previous work [20].

B. Power Amplifiers

Two three-phase linear transconductance amplifiers (LA-210 from Varedan Technologies) are used to drive the rotation and suspension windings. Each amplifier has a transconductance gain of $K_{TRANS} = 1$A/V. The bus voltage supply for the amplifiers is ±60V, and each amplifier is rated for 200 W continuous and 400 W peak output power.

C. Commutation and Control Algorithms

In this section we describe the details of the control and commutation algorithms. We first define a set of reference frames as shown in Fig. 9. A stator fixed xy-frame (black) designates the convention for positive radial forces and motion. The stator-fixed UVW-frame (green) defines the magnetic axes of the three-phase two-pole rotation winding as described in Section III-D. Similarly, the ABC-frame (purple) defines the magnetic axes of the three-phase, four-pole suspension winding. The stator-fixed $\alpha\beta$-frame (light blue) defines the magnetic axes of an equivalent two-phase representation of the three-phase suspension winding. Finally, the rotor-fixed dq-frame (orange) designates the direct and quadrature axes as discussed in Section II-B when describing the IPM torque generation principle. This frame rotates with the rotor, and the rotor angle ϕ is defined as the angle between the x-axis and the d-axis.

1) Suspension: A block diagram for the suspension control system is shown in Fig. 8a. SISO controllers, denoted as $G_{C,x}(s)$ and $G_{C,y}(s)$, are used to stabilize each radial degree of freedom. The output force commands from each controller are fed to a decoupling force commutation algorithm $C_{F,COMM}$. This algorithm computes two voltage commands $V_{A,CMD}$, $V_{B,CMD}$ which are proportional to the suspension winding phase A and B currents required to generate the commanded forces at instantaneous rotor angle $\hat{\phi}$. The voltage commands are sent to the suspension amplifier which drives the phase currents i_A and i_B through the transconductance gain K_{TRANS}. The phase current i_C is determined such that $i_A + i_B + i_C = 0$. A transformation from the ABC frame to the xy frame C_{xy}^{ABC} is included for modeling convenience.

The suspension plant dynamics ($G_{P,x}$, $G_{P,y}$) are modeled as two decoupled spring-mass systems with negative stiffness [1, pp. 43].

The rotor radial positions are measured by the optical sensors which are represented by gains $K_{sn,x}$ and $K_{sn,y}$. The position estimates are fed back and subtracted from position reference commands which are typically zero to center the rotor.

We use the decoupling force commutation algorithm

$$\begin{bmatrix} V_{A,CMD} \\ V_{B,CMD} \\ V_{C,CMD} \end{bmatrix} = C_{ABC}^{\alpha\beta} C_{\alpha\beta}^{dq} C_{dq}^{xy} \begin{bmatrix} F_{x,CMD} \\ F_{y,CMD} \end{bmatrix}, \tag{2}$$

where

$$C_{ABC}^{\alpha\beta} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \tag{3}$$

$$C_{\alpha\beta}^{dq} = \frac{1}{K_F} \begin{bmatrix} \cos\left(2\left(\phi - \frac{\pi}{12}\right)\right) & -\sin\left(2\left(\phi - \frac{\pi}{12}\right)\right) \\ \sin\left(2\left(\phi - \frac{\pi}{12}\right)\right) & \cos\left(2\left(\phi - \frac{\pi}{12}\right)\right) \end{bmatrix} \tag{4}$$

$$C_{dq}^{xy} = \begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{bmatrix} \tag{5}$$

The matrix C_{dq}^{xy} (5) is a coordinate transformation from the xy-frame to the dq-frame. The matrix $C_{\alpha\beta}^{dq}$ (4) transforms dq-frame forces to $\alpha\beta$-frame currents. This matrix implicitly contains the dq-frame force-to-current relationship, a two-pole to four-pole conversion, and a coordinate transform from the dq-frame to the $\alpha\beta$-frame. We assume the d-axis and q-axis force-to-current relationships are equivalent and use a single scaling factor K_F. We set this factor to unity in this implementation to simplify the computation. The gain is ultimately accounted for in the proportional term of the suspension controllers. The matrix $C_{ABC}^{\alpha\beta}$ (3) is a power-invariant Clarke transformation [21] that transforms the two-phase $\alpha\beta$ frame currents to three-phase ABC-frame currents.

This decoupling scheme is based on the algorithm suggested in [1, pp. 178] for surface PM machines. A more complex algorithm is suggested in [1, pp. 181] for salient pole machines with high armature reaction, but we demonstrate in Section V that our rotor enables stable levitation with the relatively simple algorithm in (2) despite the salient poles present in our rotor. Such a decentralized control is possible and effective when the plant transfer function matrix is square and close to diagonal [22].

We assume the force commutation decouples the x and y force generation sufficiently such that SISO controllers can be used to stabilize each radial degree of freedom. We implement controllers for each axis of the form

$$G_C(s) = K_P \frac{\alpha \tau s + 1}{\frac{s^2}{\omega_0^2} + \frac{2\zeta s}{\omega_0} + 1}. \tag{6}$$

The controller parameters were tuned using the frequency domain design strategy of loop shaping to achieve 200 Hz loop crossover frequency with 50 degree phase margin for the y-axis suspension dynamics with the rotor at $\phi = 0$. The final controller parameters are shown in Table I. The same parameters were used for both the x-axis and y-axis SISO controller implementations.

This continuous time controller design is converted to a discrete time equivalent with a backward Euler transform. Both the suspension control and force commutation algorithms are implemented in LabVIEW to run on the cRIO 9057 FPGA using fixed-point arithmetic. The control and commutation loop is run at 20 kHz.

2) Rotation: A block diagram of the open-loop rotation control system is shown in Fig. 8b. A simple field-oriented control scheme is used for torque commutation. The commutation algorithm $C_{\tau,COMM}$ consists of a power-invariant Clarke transformation [21] multiplied by a Park transformation [23]. This algorithm is implemented in LabVIEW to run on the cRIO 9057 FPGA using fixed-point arithmetic. The commutation loop is run at 20 kHz.

V. Test Results

In this section we present results from suspension and rotation testing of the new bearingless motor. We demonstrate stable levitation and a maximum rotation speed of 6156 RPM.

A. Suspension Tests

We performed several tests to demonstrate stable levitation and characterize the rotor-angle-dependent suspension dynamics. We characterized the suspension dynamics at two rotor angles, $\phi = 15°$ and $\phi = 45°$. Step response results at each of these rotor angles are shown in Fig. 10 and Fig. 11. For these tests, a 0.1 V (23 μm) step in position is commanded for each axis independently. For all tests, some level of force cross-coupling is present as indicated by the non-zero cross-axis responses. However, the coupling is small enough to maintain stable levitation.

The tabulated rise times and overshoots computed from the step response tests are shown in Table II. For the $\phi = 15°$ position, the y-axis response has a faster rise time and more overshoot than the x-axis response. In the $\phi = 45°$ position, both rise times and overshoots for x and y are similar. These observations are consistent with the predicted rotor-angle-dependent suspension dynamics derived from the simulated parameters presented in Section III-A.

Frequency response measurements were also performed to characterize the suspension dynamics. The loop return ratio was measured at the two rotor positions using a dynamic signal analyzer (DSA) implemented on the cRIO 9057 real-time processor running at 1 kHz. The measured responses are shown in Fig. 12, and the associated loop crossover frequencies and phase margins are tabulated in Table III. The measured phase margin for all cases is less than the targeted 50°. We attribute this in part to the delay in the DSA implementation, as the step responses indicate the system is more damped than indicated by the measured phase margin. The unity gain crossover frequency varies for each case due to the rotor-angle-dependent parameter variation.

We also performed tests to characterize the dynamics of the passively stable degrees of freedom. Fig. 13 shows the natural responses to perturbations in the axial and tilt directions while the rotor is levitated and not rotating. To perturb the rotor, it was manually offset using a finger in the degree of freedom to be excited and then released. The tilt and axial position estimates are produced from the Hall element signals as discussed in Section IV-A. The tilts were excited and measured about the rotor fixed *dq*-axes. From these responses, the axial and tilting stiffnesses are calculated to be $K_{p,z}$ = 3.9N/mm, $K_{p,\theta_d} = 7.78\text{mNm}/°$, and $K_{p,\theta_q} = 27.2\text{mNm}/°$. The large asymmetry in the tilt stiffnesses may be problematic if these degrees of freedom are excited during pumping applications. Improving this symmetry will be investigated in the future.

B. Rotation Tests

Open-loop rotation tests were performed to evaluate and characterize the torque performance. For the tests, the rotor is levitated and various levels of *q*-axis current are commanded. The results from this test are shown in Fig. 14. A maximum speed of 6156 RPM was attained with i_q = 1.6A. At higher speeds, the suspension loop becomes unstable and the rotor touches down. We suspect the instability is caused by saturation of the suspension loop control effort.

We also observed that the motor has high cogging torque, which is consistent with the finite element simulations presented in Section III-A. When the rotor is levitated and not spinning, it rests at one of 12 cogging positions aligned with the stator teeth. It takes a minimum *q*-axis current command of 0.5 A to overcome the initial cogging torque when the rotor is stationary. We attribute the high cogging torque to the strong permanent magnets and the stator saliency.

VI. Conclusions

In this paper, we have presented a new segmented dipole IPM bearingless slice rotor that exhibits relatively symmetric suspension dynamics, exhibits a reduced trade-off between force and torque capacity, and is simple to manufacture. Through testing we demonstrate stable suspension and open-loop rotation using simple control and commutation algorithms.

We also identify several issues with the motor that need to be addressed in future iterations. These include the high cogging torque and the asymmetric passive dynamics. Given the initial demonstrated performance, the segmented dipole IPM slice motor shows promise for ECLS applications as well as other applications which require a non-contact drive solution. The Hall element-based sensing system also shows promise for future use in prototype bearingless motor systems to provide both angular position estimates and diagnostic estimates of the rotor tilt and axial motions. Future work will focus on optimization of the rotor geometry to improve the symmetry of the passive suspension dynamics and reduce the cogging torque.

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References

[1]. Chiba A, Fukao T, Ichikawa O, Oshima M, Takemoto M, and Dorrell DG, Magnetic Bearings and Bearingless Drives. Amsterdam, The Netherlands: Elvesier-Newnes, 2005.

[2]. Schöb R and Barletta N, "Principle and application of a bearingless slice motor," in Proc. the 5th Int. Sym. on Magn. Bearings, 8. 1996, pp. 313–318.

[3]. Silber S, Amrhein W, Bösch P, Schöb R, and Barletta N, "Design aspects of bearingless slice motors," IEEE/ASME Trans. on Mechatronics, vol. 10, no. 6, pp. 611–617, 12. 2005.

[4]. Barletta N and Schöb R, "Design of a bearingless blood pump," in Proc. 3rd Int. Symp. Magn. Susp.Technol, 7. 1996, pp. 265–274.

[5]. Nussbaumer T, Karutz P, Zurcher F, and Kolar JW, "Magnetically levitated slice motors – an overview," IEEE Trans. on Ind. App, vol. 47, no. 2, pp. 754–766, 3. 2011.

[6]. Gruber W and Silber S, "20 years bearingless slice motor – its developments and applications," in Proc. the 15th Int. Symp. on Magnetic Bearings, 8. 2016, pp. 91–98.

[7]. Mitterhofer H, Gruber W, and Amrhein W, "On the high speed capacity of bearingless drives," IEEE Trans. on Ind. Elect, vol. 61, no. 6, pp. 3119–3126, 6. 2014.

[8]. Asama J, Kanehara D, Oiwa T, and Chiba A, "Development of a compact centrifugal pump with a two-axis actively positioned consequent-pole bearingless motor," IEEE Trans. on Ind. App, vol. 50, no. 1, pp. 288–295, 1. 2014.

[9]. Gruber W, Rothböck M, and Schöb R, "Design of a novel homopolar bearingless slice motor with reluctance rotor," IEEE Trans. on Ind. App, vol. 51, no. 2, pp. 1456–1464, 3. 2015.

[10]. Noh M and Trumper DL, "Homopolar Bearingless Slice Motor With Flux-Biasing Halbach Arrays," IEEE Trans. on Ind. Elec, vol. 67, no. 9, pp. 7757–7766, 9. 2020.

[11]. Noh M, "Homopolar bearingless slice motors with magnet-free rotors for extracorporeal life support," Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA, 2018.

[12]. Noh M, Gruber W, and Trumper DL, "Hysteresis bearingless slice motors with homopolar flux-biasing," IEEE/ASME Trans. on Mechatronics, vol. 22, no. 5, pp. 2308–2318, 10. 2017. [PubMed: 29503520]

[13]. Holenstein T, Schuck M, and Kolar JW, "Performance benchmarking of a novel magnet-free bearingless synchronous reluctance slice motor," IEEE Open Journ. of the Ind. Elec. Society, vol. 1, pp. 184–193, 2020.

[14]. Turk N, Bulić N, and Gruber W, "Nonlinear Control of a Bearingless Flux-Switching Slice Motor With Combined Winding System," IEEE/ASME Trans. on Mechatronics, vol. 25, no. 1, pp. 152–163, 2. 2020.

[15]. Weinreb B, “A novel magnetically levitated interior permanent magnet slice motor,” Master’s thesis, Massachusetts Institute of Technology, Cambridge, MA, 2020.

[16]. Ooshima M, Chiba A, Rahman A, and Fukao T, “An improved control method of buried-type ipm bearingless motors considering magnetic saturation and magnetic pull variation.”

[17]. Takemoto M, Uyama M, Chiba A, Akagi H, and Fukao T, “A deeply-buried permanent magnet bearingless motor with 2-pole motor windings and 4-pole suspension windings,” in Proc. 38th IAS Annual Meeting on Con. Rec. of the Ind. App. Conf, 10. 2003, pp. 1413–1420.

[18]. Lovelace E, “Optimization of a magnetically saturable interior permanent-magnet synchronous machine drive,” Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA, 2000.

[19]. Meeker D, Finite Element Method Magnetics, User’s Manual Version 4.2, 2007. [Online]. Available: http://www.femm.info/wiki/HomePage

[20]. Weinreb B, Fyler D, and Trumper D, “A novel position sensing system for bearingless motors,” in Proc. 2020 ASPE Spring Topical Meeting–Design and Control of Precision Mechatronic Systems, 5 2020, pp. 37–42.

[21]. Duesterhoeft W, Schulz M, and Clarke E, “Determination of Instantaneous Currents and Voltages by Means of Alpha, Beta, and Zero Components,” Trans. of the American Inst. of Elec. Engineers, vol. 70, no. 2, pp. 1248–1255, 7. 1951.

[22]. Skogestad S and Postlethwaite I, Multivariable feedback control: analysis and design, 2nd ed. Chichester, England: Wiley, 2005.

[23]. Park RH, “Two-reaction theory of synchronous machines generalized method of analysis-part I,” Trans. of the American Inst. of Elec. Engineers, vol. 48, no. 3, pp. 716–727, 7. 1929.

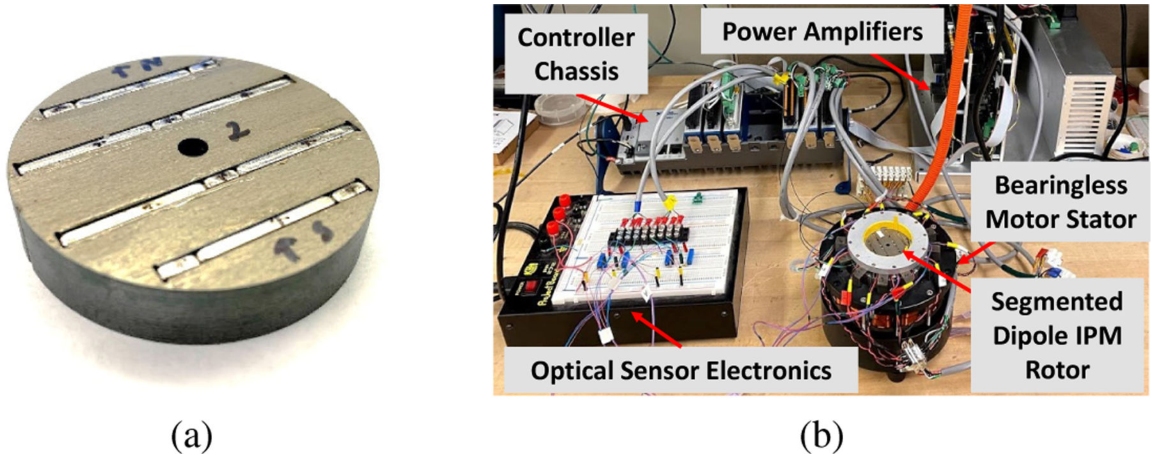

Fig. 1.

Photographs of the (a) prototype segmented dipole IPM rotor and (b) prototype bearingless motor system.

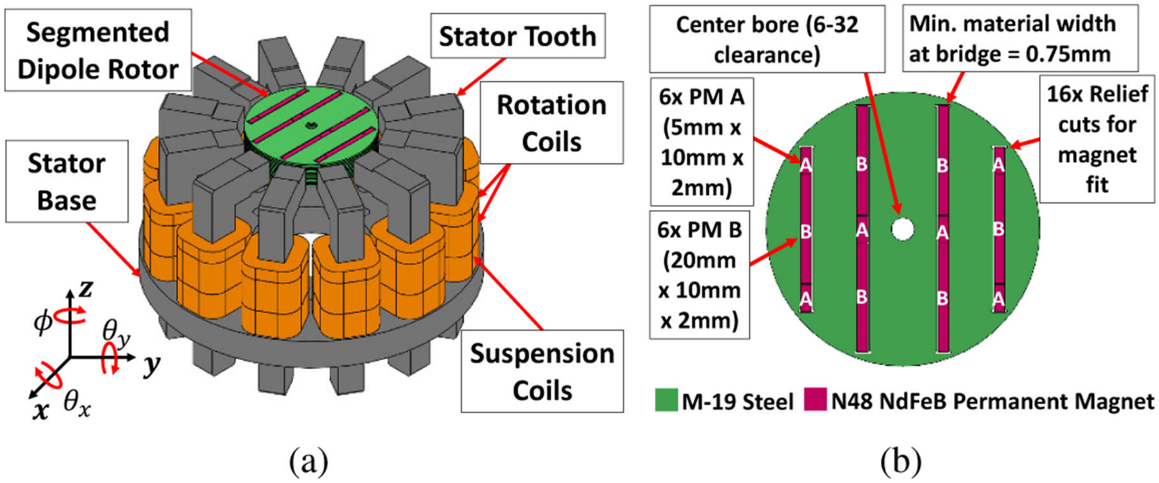

Fig. 2.

Renderings of the prototype bearingless motor stator highlighting the (a) rotor, stator structure, and coils and (b) the details of the rotor construction.

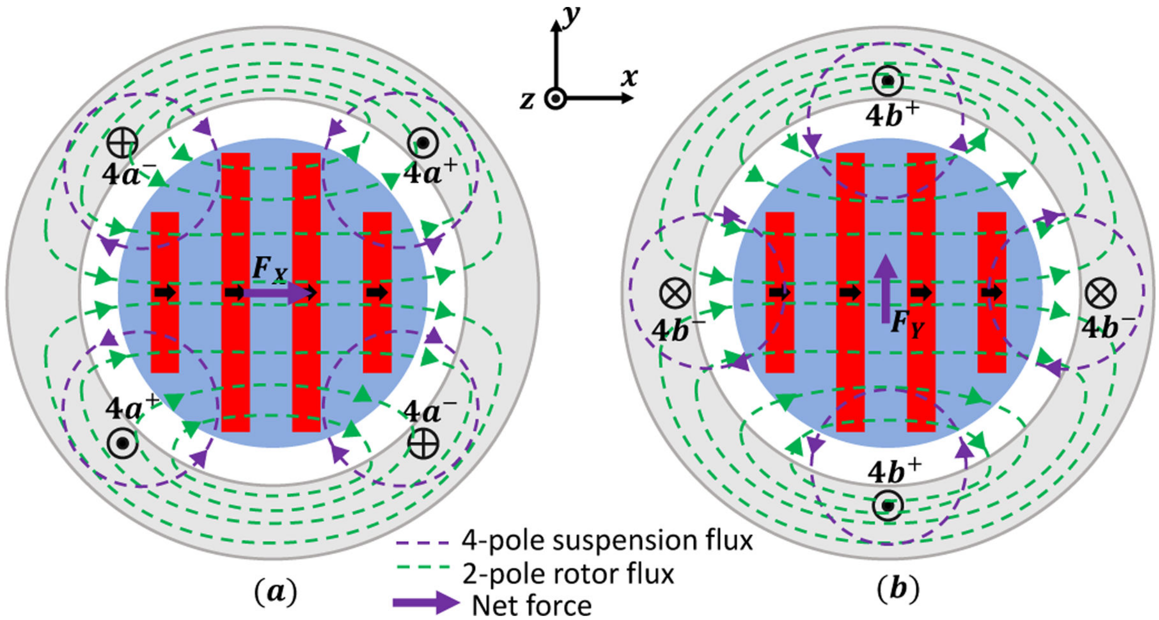

Fig. 3.
Illustration of the (a) x force generation and (b) y force generation in the segmented dipole IPM rotor. In (a), the net flux density is increased on the right side of the rotor and decreased on the left side, producing a net x-force on the rotor. In (b), the net flux density is increased on the top side of the rotor and decreased on the bottom side, producing a net y-force on the rotor.

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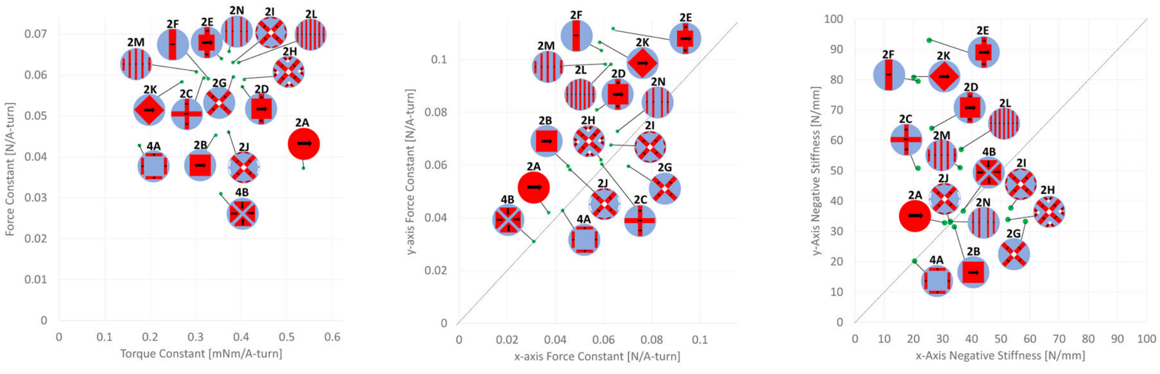
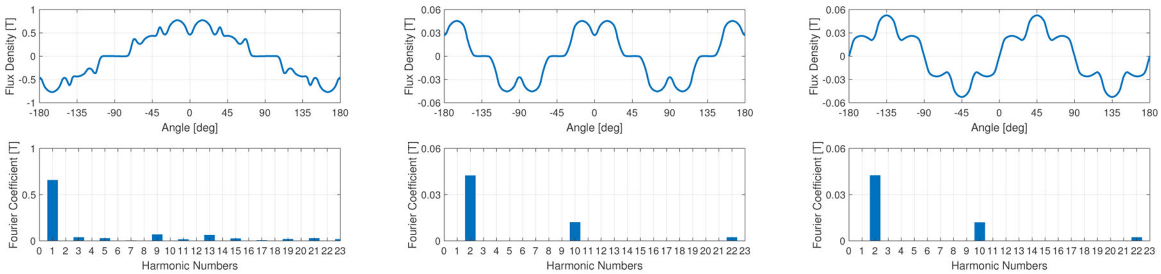


(a) (b) (c)

Fig. 4.

(a) Chart comparing the simulated force and torque constants of several candidate rotor designs. The force constant plotted is the minimum of the simulated x and y force constants. (b) Chart comparing the simulated x and y force constants of several candidate rotor designs. (c) Chart comparing the simulated x and y negative stiffnesses of several candidate rotor designs. All simulations were performed with the rotor angle $\phi = 0$.



(a) (b) (c)

Fig. 5.

Plots of the simulated air-gap magnetic flux density distributions for the segmented dipole IPM rotor design (2N) with mechanical angle $\phi = 0$. (a) Air-gap magnetic flux distribution and FFT produced by the permanent magnets. (b) Air-gap magnetic flux distribution and FFT produced when the x-axis suspension windings are excited with 1A current and the magnets are replaced with air. (c) Air -gap magnetic flux distribution and FFT produced when the y-axis suspension windings are excited with 1A current and the magnets are replaced with air.

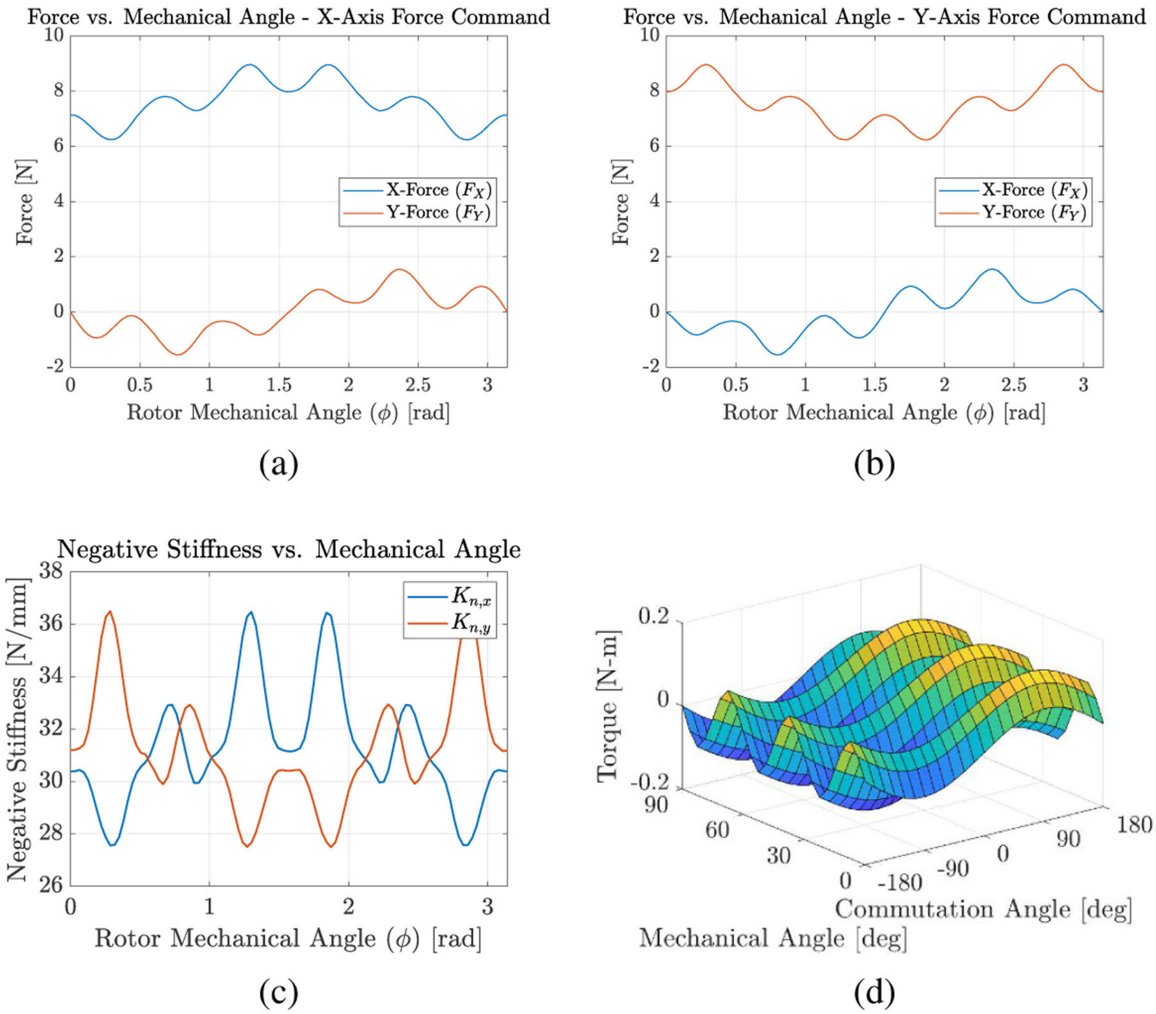


Fig. 6.

Plots of the simulated angle dependent performance parameters for the segmented dipole IPM rotor: (a) Radial forces vs. rotor mechanical angle for 1A x-axis current command. (b) Radial forces vs. rotor mechanical angle for unity y-axis current command. (c) Negative stiffness vs. rotor mechanical angle. (d) Torque vs. mechanical angle vs. commutation angle (torque ripple).

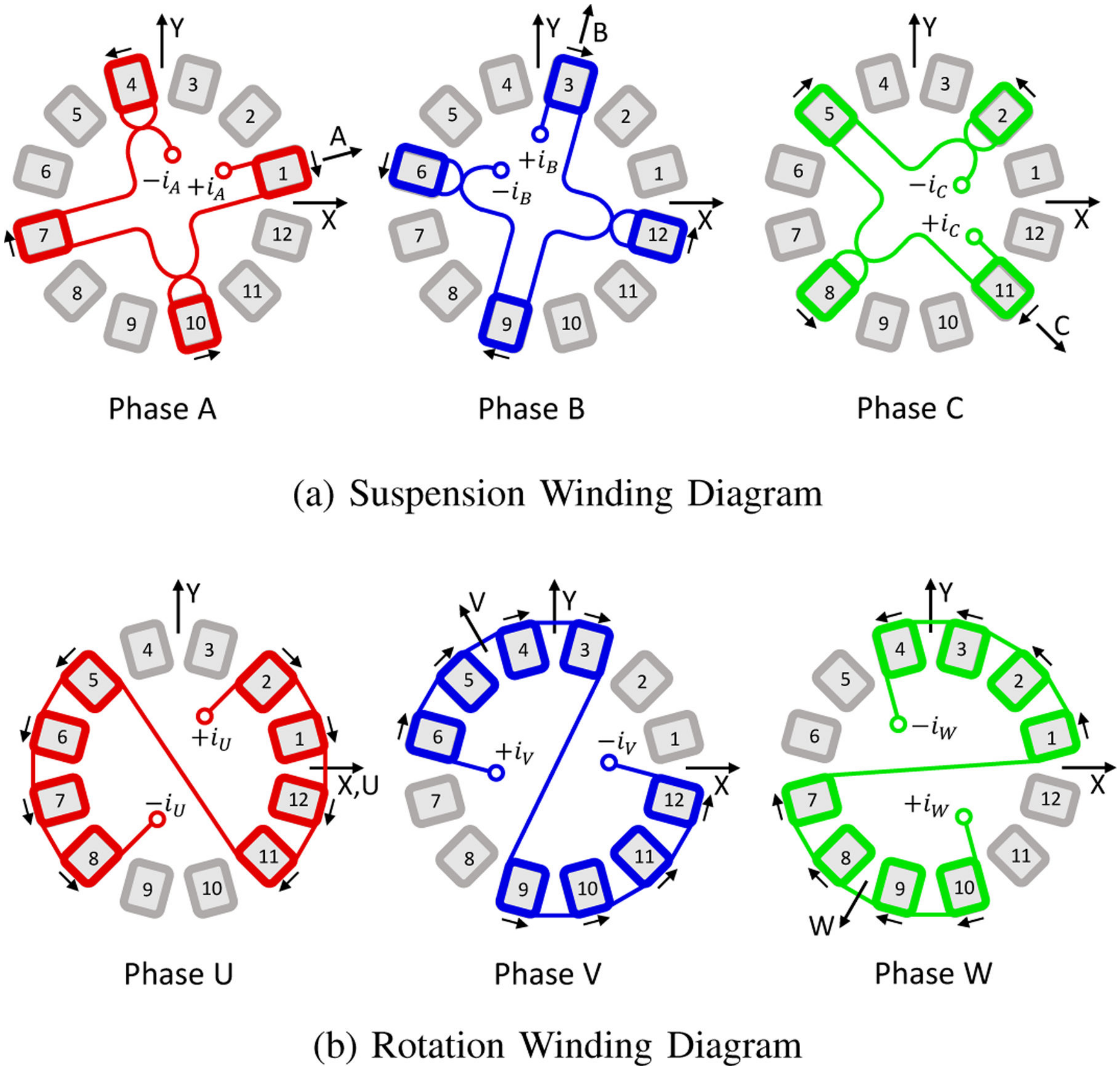
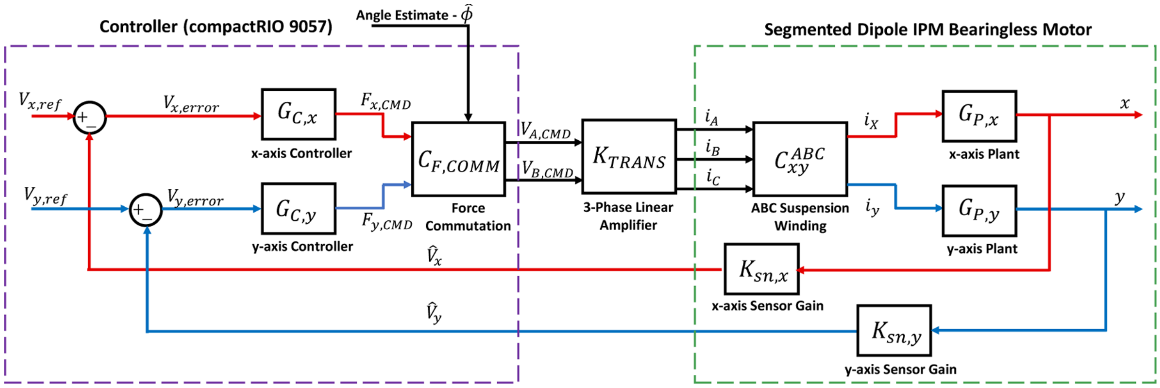
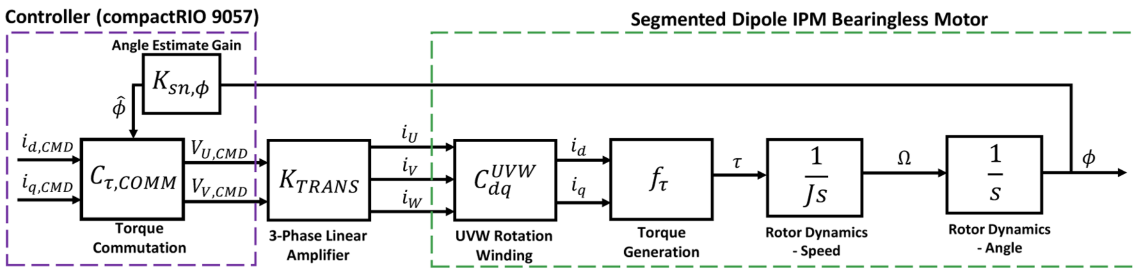


Fig. 7.

Diagrams of the rotation and suspension windings.



(a) Suspension Block Diagram



(b) Rotation Block Diagram

Fig. 8.

Block diagrams of the control and commutation schemes for both rotation and suspension.

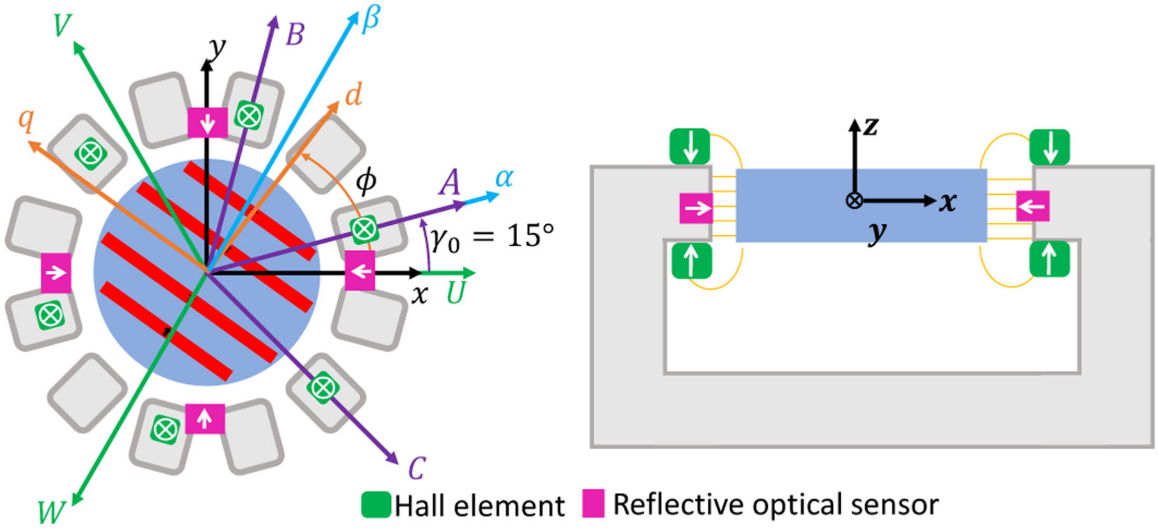


Fig. 9.
Configuration of the position and angle sensors in the bearingless motor prototype. The white arrows on the sensors denote the sensitive axes. Also shown on the left are the reference frame conventions for control and commutation.

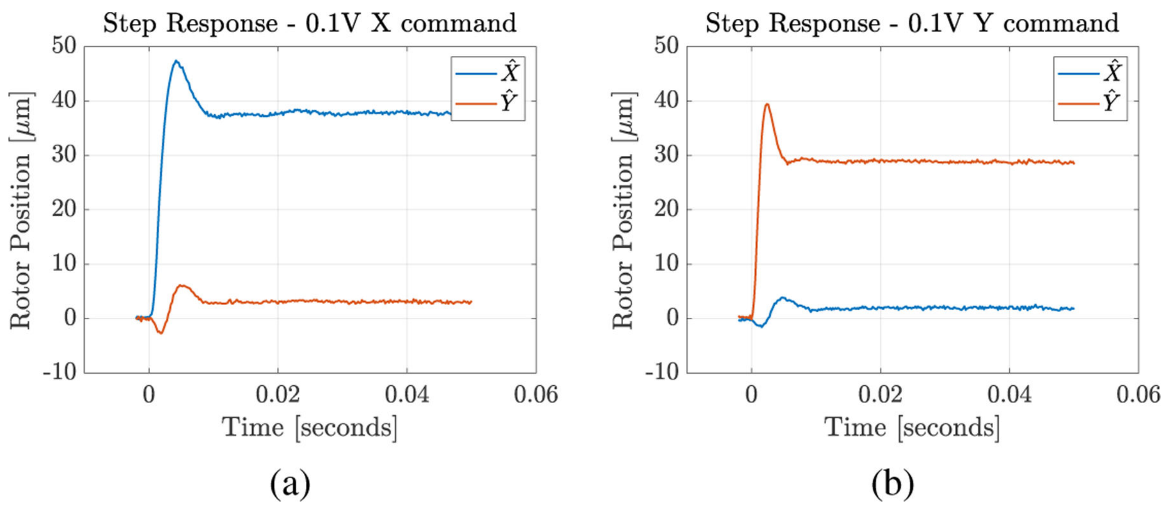


Fig. 10.

Measured radial position data for a 0.1 V commanded step in (a) x-position and (b) y-position with the rotor oriented at $\phi = 15°$. The cross axis responses (ideally zero) are also shown.

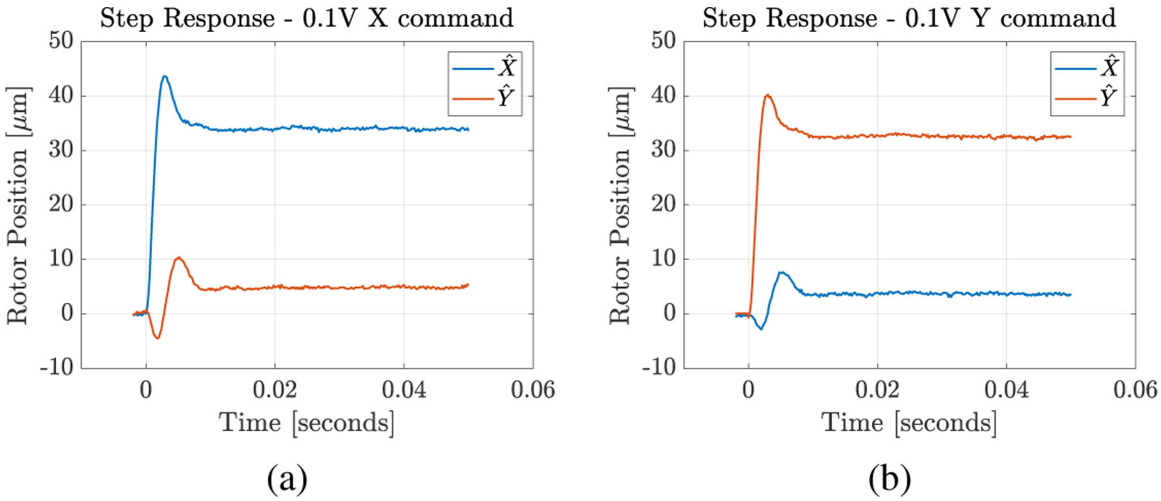


Fig. 11.

Measured radial position data for a 0.1 V commanded step in (a) x-position and (b) y-position with the rotor oriented at $\phi = 45°$. The cross axis responses (ideally zero) are also shown.

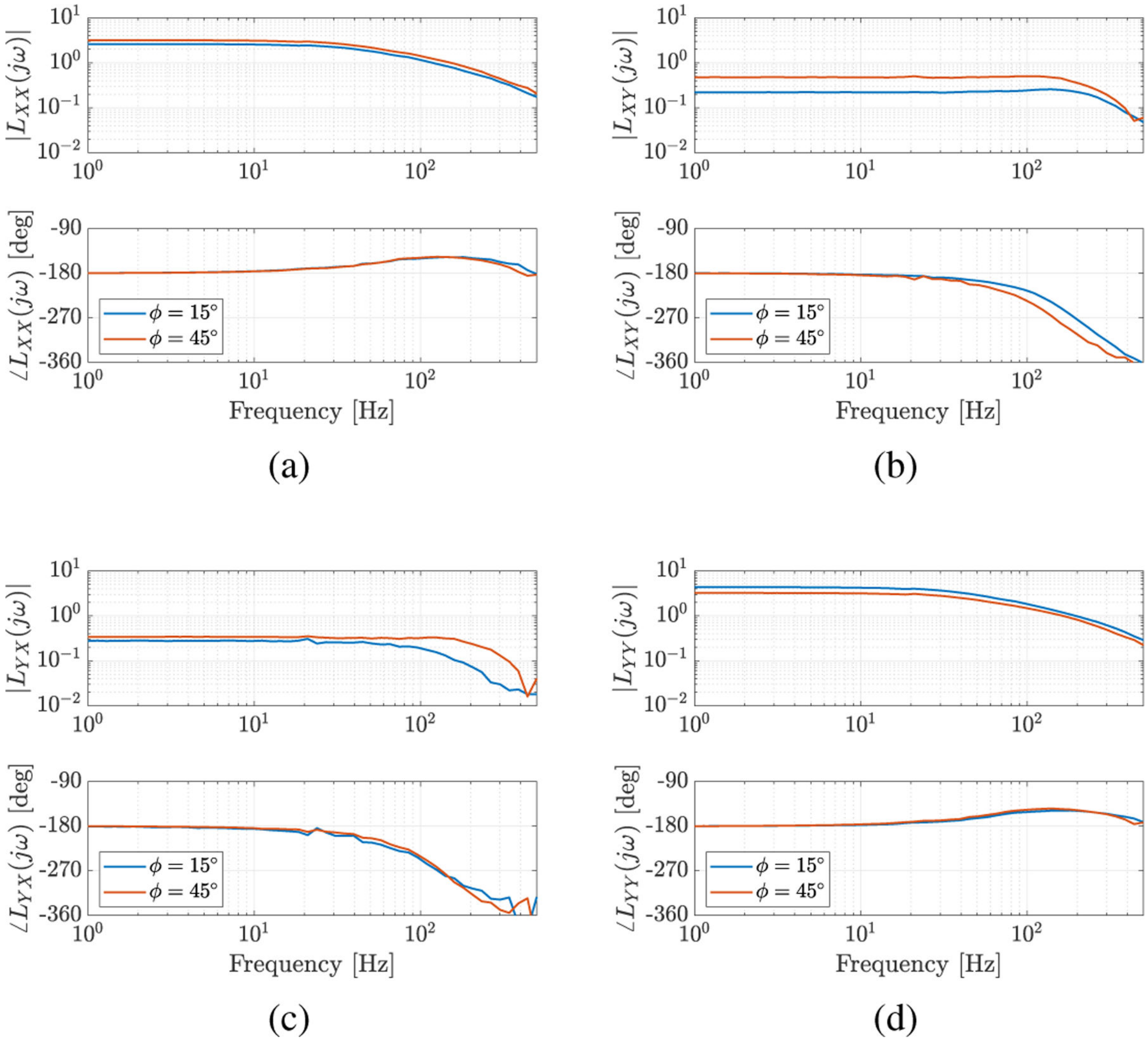


Fig. 12.

Measured loop return ratio frequency responses with the rotor oriented at ϕ = 15° (blue) and ϕ = 45° (red). Plots (a) and (d) show the measured on-axis (x and y) loop return ratios. Plots (b) and (c) show the measured cross-axis loop return ratios.

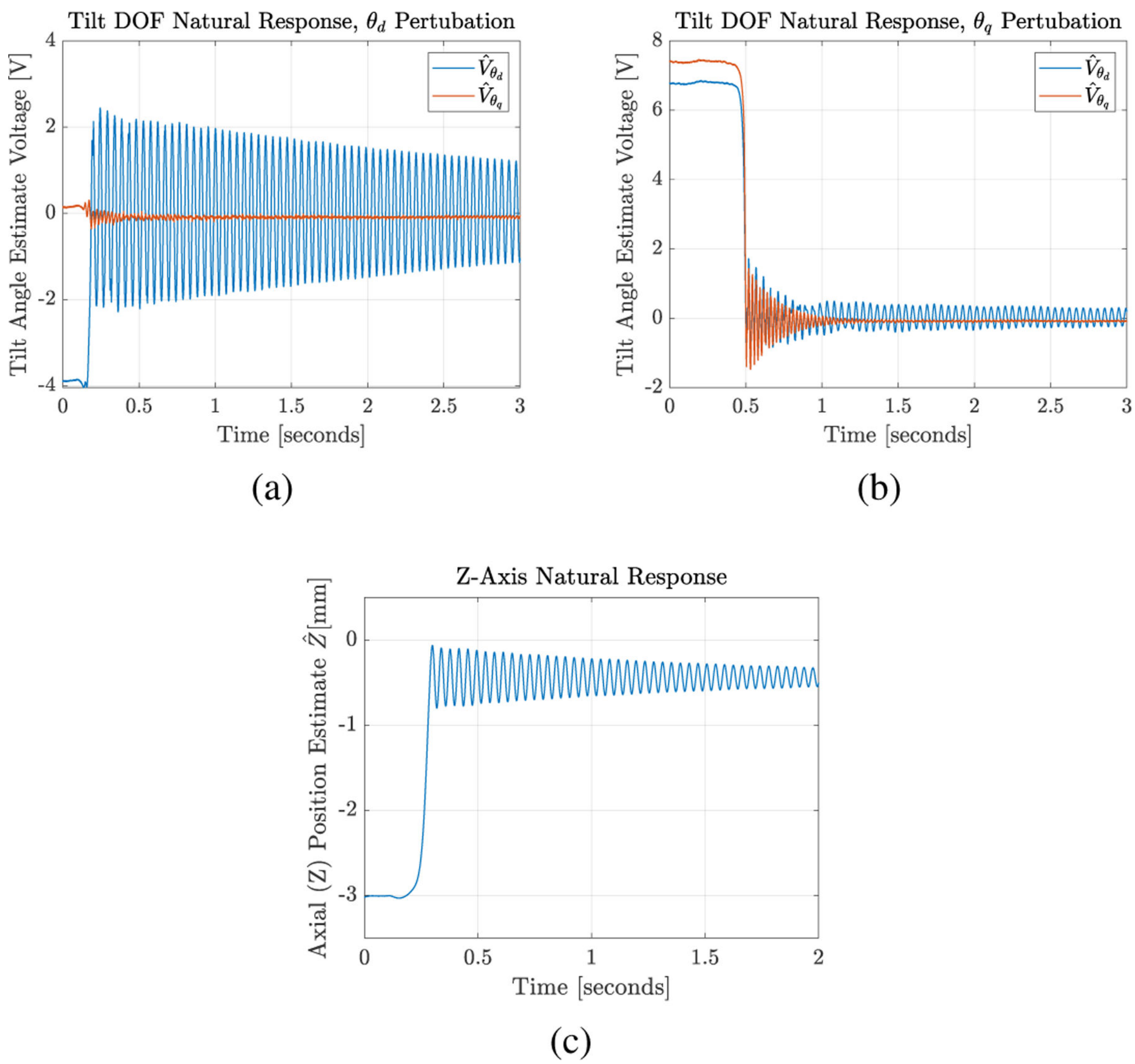


Fig. 13.

Measured natural responses to a (a) *d*-axis tilt perturbation, (b) *q*-axis tilt perturbation, and (c) z-axis translational perturbation at rotor angle $\phi = 15°$. The tilt estimates and axial estimates are computed from the Hall element signals. The cross-axis responses for the tilt perturbations are also shown. Note the lightly damped responses.

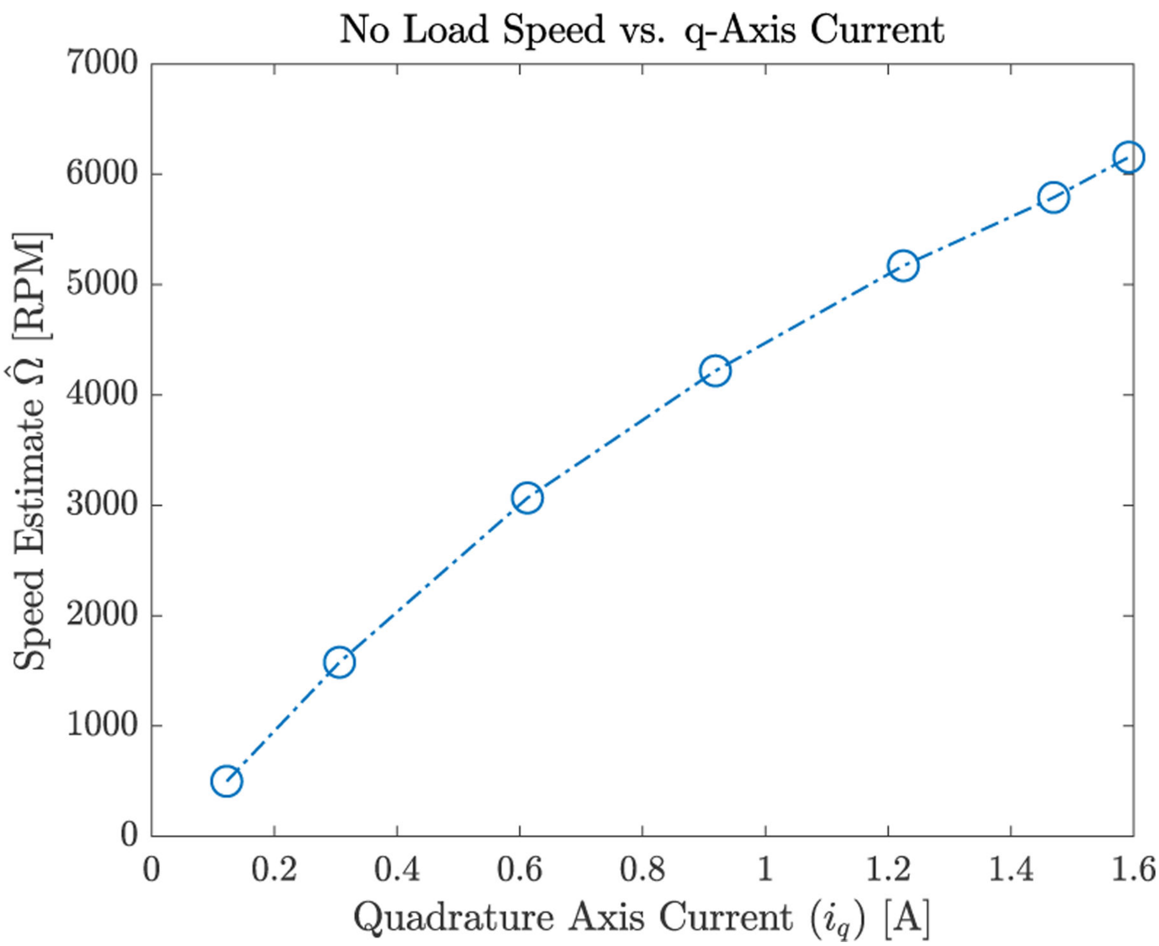


Fig. 14.

Plot of open loop rotation experiment results. Commanded q-axis current vs. measured no load speed.

Table I

Suspension Controller Parameters

Parameter	Value
K_p	2.4418
a	6
τ	$3.242\,49 \times 10^{-4}$ sec
ω_0	6156 rad/sec
ζ	0.7071 V/m

Table II

Measured step response parameters

Test Case	Rise Time (10%−90%)	Percent Overshoot
x, $\phi = 15°$	2.3 ms	25.5%
y, $\phi = 15°$	1.3 ms	38.9%
x, $\phi = 45°$	1.6 ms	29.8%
y, $\phi = 45°$	1.6 ms	23.4%

Table III

Measured frequency response parameters

Test Case	Loop Crossover Frequency	Phase Margin
L_{xx}, $\phi = 15°$	117 Hz	31°
L_{yy}, $\phi = 15°$	194 Hz	31°
L_{xx}, $\phi = 45°$	153 Hz	32°
L_{yy}, $\phi = 45°$	160 Hz	35°